Problem Summary

You are climbing a staircase with n steps.

Each time, you can climb:

→ 1 step

→ 2 step

Return the number of distinct ways to reach the top.

Key Idea

To reach step n, there are only 2 possible previous places:

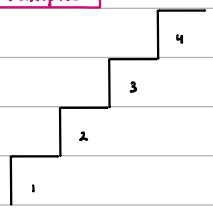
→ from step n-1 by taking 1 step

→ from step n-2 by taking 2 steps

So: ways(n) = ways(n-1) + ways(n-2) ⟶ Same pattern as Fibonacci

Diagram Examples

n = 4



ways(5) = ways(4) + ways(3)

= [ways(3) + ways(2)] + [ways(2) + ways(1)]

= [(ways(2) + ways(1)) + ways(2)] + [ways(2) + ways(1)]

= 2 + 1 + 2 + 2 + 1

= 8

Approach

1) Handle base cases:

→ n=1 : return 1

→ n=2 : return 2

2) Set:

→ prev = 1

→ curr = 2

3) For each step from 3 to n: update prev and curr

4) Return curr

Time Complexity: O(n) → we loop through the steps once

Space Complexity: O(1) → only two variables used